LAB 6: Draw the Sequence Diagram

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Project: AI Chatbot for University Helpdesk

1. Sequence Diagram: Student Asks a Question (RAG Pipeline)

Shows the message flow between the student, React frontend, FastAPI backend, authentication service, RAG service, ChromaDB, the LLM API and SQLite. An alt fragment covers the case where no relevant context is found and the question is logged as unanswered. The feedback step follows the answer.

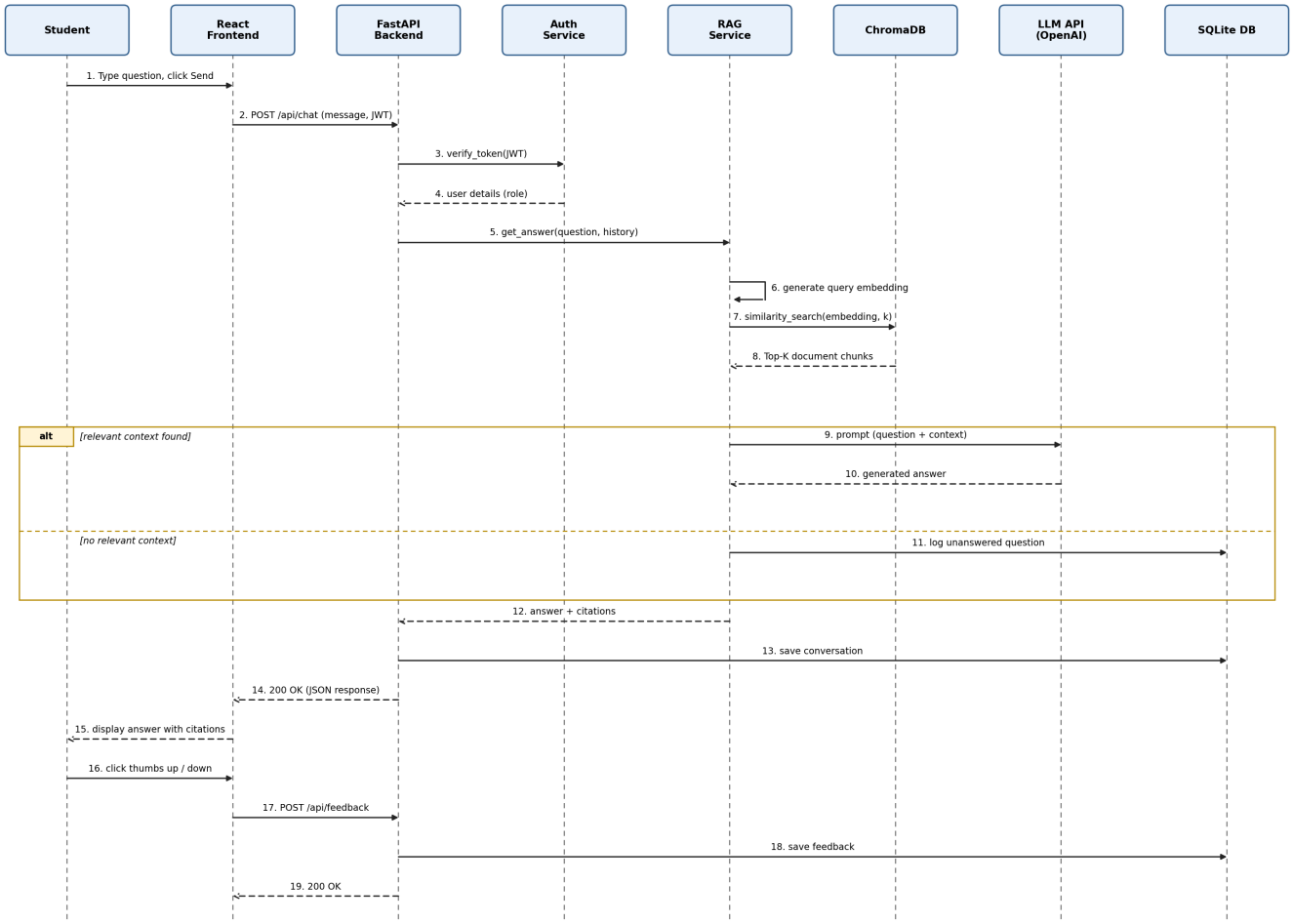


Figure 6.1: Sequence diagram for chat query and feedback

2. Sequence Diagram: Administrator Uploads a Document

Shows the document ingestion flow: the token is verified, then the text is extracted, chunked and embedded, vectors are stored in ChromaDB, and metadata is saved in SQLite. Non-admin users receive a 403 response.

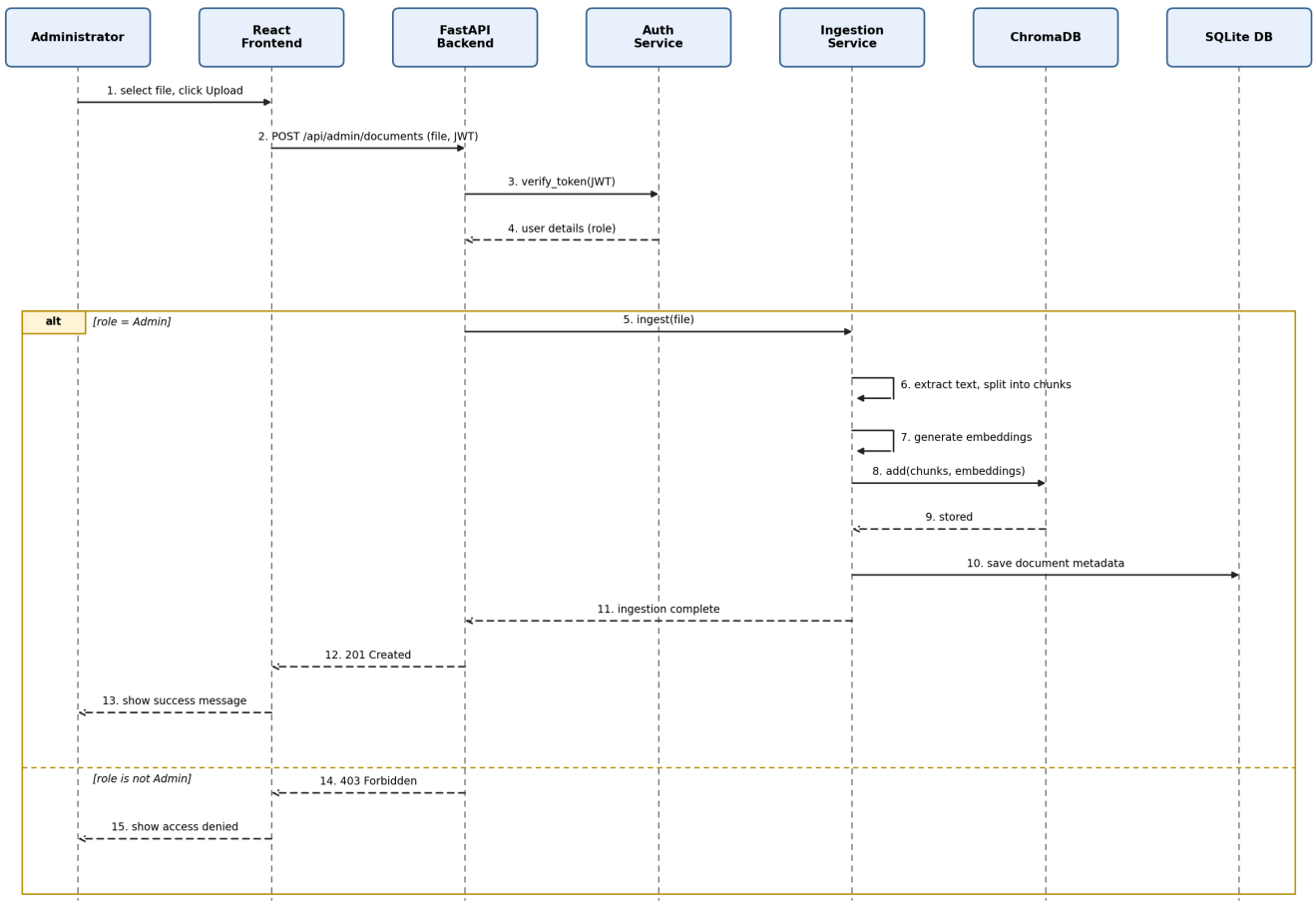


Figure 6.2: Sequence diagram for document upload